Q1.

(a) When fertilisation takes place, the nucleus of a sperm joins with the nucleus of an ovum (egg).

In which part of the reproductive system does fertilisation normally take place in humans?
Tick the correct box.

cervix ☐ ovary ☐

oviduct ☐ uterus ☐

1 mark

(b) The table below gives information about fertilisation in three animals.

animal	Does fertilisation take place inside or outside the body?	Number of eggs released at a time
human	inside	1
bird	inside	4
frog	outside	3000

Frogs release their eggs and sperm into water.
The eggs are fertilised in the water.

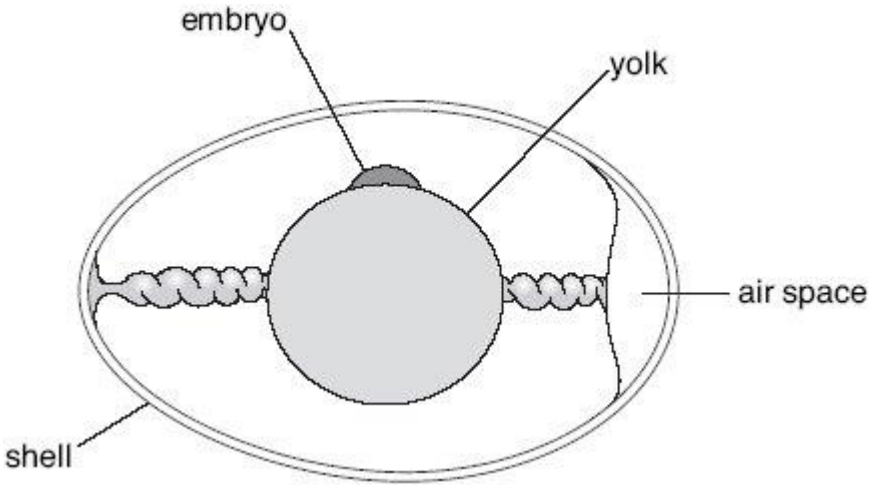
Why is it an advantage for frogs to release large numbers of eggs and sperm?

.....

.....

1 mark

(c) The diagram shows a section through a fertilised egg of a bird.



(i) The shell of a bird's egg is porous. This means it has microscopic holes in it.

Why does it need to be porous?

.....

1 mark

(ii) Give **one** other function of the egg shell.

.....

1 mark

(d) A bird's egg contains yolk which is a food store for the developing chick. A human egg does **not** contain yolk.

Why does a human egg **not** need to contain a food store for the embryo?

.....

.....

1 mark

maximum 5 marks

Q1.

(a) oviduct ✓

if more than one box is ticked, award no mark

1 (L6)

(b) any **one** from

- some eggs **or** sperm might be eaten **or** lost
- eggs **or** sperm are not protected inside the body
- to increase the chance of an egg being fertilised
  accept 'to make fertilisation more likely'
- some eggs might not be fertilised
- some are washed away
  accept 'some die'
- some of the offspring might be eaten
  accept 'more tadpoles'

1 (L5)

(c) (i) any **one** from

- to let oxygen **or** air into the egg
- to let carbon dioxide **or** waste gases out
- to allow gas exchange
  accept 'for respiration'
  *do **not** accept 'to let waste out'*

1 (L5)

(ii) any **one** from

- for protection
- to stop the egg drying out
  accept 'it stops it drying out'
- it stops infection **or** disease
  *accept 'it stops micro-organisms **or** bacteria **or** viruses **or** germs getting in'*
  accept 'to retain the contents'
  'it stops infection getting in' is insufficient

1 (L5)

(d) any **one** from

- food passes from the mother to the embryo **or** baby
  accept 'the mother provides food'

- the embryo is fed through the placenta
  accept 'it is fed through the umbilical cord'

- it gets food from the mother's blood
  *do **not** accept 'it gets food from the blood'*

1 (L6)

[5]